Computer Lab: SQL Server-8

Part 2: Data Manipulation(continue)

7.8 Database Updates

In this section, we describe the three SQL statements that are available to modify the contents of the tables in the database:

- INSERT – adds new rows of data to a table;
- UPDATE – modifies existing data in a table;
- DELETE – removes rows of data from a table.

Adding data to the database (INSERT)

There are two forms of the INSERT statement. The first allows a single row to be inserted into a named table and has the following format:

```
INSERT INTO TableName [(columnList)]
VALUES (dataValueList)
```

TableName may be either a base table or an updatable view, and *columnList* represents a list of one or more column names separated by commas. The *columnList* is optional; if omitted, SQL assumes a list of all columns in their original CREATE TABLE order. If specified, then any columns that are omitted from the list must have been declared as NULL columns when the table was created, unless the DEFAULT option was used when creating the column. The *dataValueList* must match the *columnList* as follows:

- the number of items in each list must be the same;
- there must be a direct correspondence in the position of items in the two lists, so that the first item in *dataValueList* applies to the first item in *columnList*, the second item in *dataValueList* applies to the second item in *columnList*, and so on;
- the data type of each item in *dataValueList* must be compatible with the data type of the corresponding column.

Example 7.35 INSERT . . . VALUES

Insert a new row into the Staff *table supplying data for all columns.*

```
INSERT INTO Staff
VALUES ('S5', 'Farid', 'Saleh', 'Mansoura', 'Assistant',8300, 'B03');
```

As we are inserting data into each column in the order the table was created, there is no need to specify a column list. Note that character literals such as 'Farid' must be enclosed in single quotes.

Staff

staffNO	fName	lName	city	position	salary	branchNo
s1	Gad	Ahmed	Tanta	Manager	60,000	B02
s2	Samy	Tamer	Cairo	Assistant	30,000	B02
s3	Samy	Ahmed	Tanta	Supervisor	50,000	B01
s4	Tamer	Samir	Giza	Manager	40,000	B03

Example 7.36 INSERT using defaults -insert data in specific columns

Insert a new row into the Staff *table supplying data for the following columns:* staffNo*,* fName*,* lName*,* position*,* salary*, and* branchNo*.*

INSERT INTO Staff (staffNo, fName, lName, position, salary, branchNo)
VALUES ('S6', 'Anne', 'Jones', 'Assistant', 8100, 'B03');

In this case, we have explicitly specified that the column city should be set to NULL.

The second form of the INSERT statement allows multiple rows to be copied from one or more tables to another, and has the following format:

INSERT INTO TableName [(columnList)]
SELECT . . .

TableName and *columnList* are defined as before when inserting a single row. The SELECT clause can be any valid SELECT statement. The rows inserted into the named table are identical to the result table produced by the subselect. The same restrictions that apply to the first form of the INSERT statement also apply here.

Example 5.37 INSERT . . . SELECT

Assume that there is a table StaffProject that contains the names of staff and the projects they manage:

StaffProject (staffNo, fName, lName, projNo)

Populate the StaffProject *table using details from the* Staff *and* Project *tables.*

```
INSERT INTO StaffProject
(SELECT s.staffNo, fName, lName, projNo
FROM Staff s, Project p
WHERE s.staffNo = p.staffNo;
```

The StaffProject table will now be as shown in Table 7.35.

Table 7.35 Result table for Example 7.37.

staffNO	fName	lName	projNo
s1	Gad	Ahmed	PT21
s2	Samy	Tamer	PT32
s3	Samy	Ahmed	PT11

هعمل جدول جديد في الداتا بيز يكون فيه بيانات الربط ما بين جدولين معمول لهم جوين
هدخل فيه قيم جديده
القيم ده جايه من ربط الجدولين

Modifying data in the database (UPDATE)

The UPDATE statement allows the contents of existing rows in a named table to be changed. The format of the command is:

```
UPDATE TableName
SET columnName1 = dataValue1 [, columnName2 = dataValue2 . . . ]
[WHERE searchCondition]
```

TableName can be the name of a base table or an updatable view. The SET clause specifies the names of one or more columns that are to be updated. The WHERE clause is optional; if omitted, the named columns are updated for *all* rows in the table. If a WHERE clause is specified; only those rows that satisfy the *searchCondition* are updated. The new *dataValue*(*s*) must be compatible with the data type(s) for the corresponding column(s).

Example 7.38 UPDATE all rows

Give all staff a 3% pay increase.

```
UPDATE Staff
SET salary = salary*1.03;
```

As the update applies to all rows in the Staff table, the WHERE clause is omitted.

Example 7.39 UPDATE specific rows

Give all Managers a 5% pay increase.

```
UPDATE Staff
SET salary = salary*1.05
WHERE position = 'Manager';
```

The WHERE clause finds the rows that contain data for Managers and the update salary = salary*1.05 is applied only to these particular rows.

Example 7.40 UPDATE multiple columns

*Promote Samer Tamer (*staffNo = *'S2') to Manager and change his salary to £60,000.*

```
UPDATE Staff
SET position = 'Manager', salary = 60000
WHERE staffNo = 'S2';
```

Deleting data from the database (DELETE)

The DELETE statement allows rows to be deleted from a named table. The format of the command is:

```
DELETE FROM TableName
 [WHERE searchCondition]
```

As with the INSERT and UPDATE statements, *TableName* can be the name of a base table or an updatable view. The *searchCondition* is optional; if omitted, *all* rows are deleted from the table. This does not delete the table itself – to delete the table contents and the table definition, the DROP TABLE statement must be used instead. If a *searchCondition* is specified, only those rows that satisfy the condition are deleted.

Example 7.41 DELETE specific rows
Delete all staff in branch B02.

```
DELETE FROM staff
WHERE branchNo = 'B02';
```

The WHERE clause finds the rows for branchNo = 'B02' and the delete operation is applied only to these particular rows.

Example 7.42 DELETE all rows
Delete all rows from the Viewing *table.*

DELETE FROM Staff;

This removes all rows from the table leaving only the table definition, so that we are still able to insert data into the table at a later stage.